

Fig. 3: An actual-size, single-side PCB for the circuit of AVR programmer using AVR910 protocol (Fig. 2)

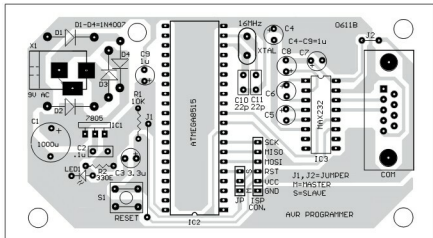


Fig. 4: Component layout for the PCB

mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC1). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and resistor R2 limits the current through LED1.

An actual-size, single-side PCB for the circuit of AVR programmer using AVR910 protocol (Fig. 2) is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

The target AVR microcontroller enters serial programming mode only when its reset pin is active-'low'. Immediately after the reset pin becomes low, serial clock (SCK) pin is driven to zero by the programmer. The reset pin makes the device SPI slave and the target device ready to accept instructions from the programmer through MOSI (master out, serial data in) and MISO (master in, serial out) pins.

The serial programming instruction set can be referred through any AVR datasheet. It can be implemented on any AVR board by just making the device at 19,200 baud rate for UART and selecting a proper pin on the master device which will make the reset pin of the slave device low, SPI

slave select input PB4 (SS) pin being the most convenient one. Three-pin connectors are provided on the programmer board for burning the code into the programmer or the target microcontroller.

To program the ATmega8515 on the programmer board, connect a 6-wire cable between ISP connectors (marked as ISP6PIN) of the STK500 board and the programmer board. Connect a serial cable from the connector marked as 'RS232 CTRL' on the STK500 board to COM port of the PC. Connect the reset pin (RST) of the ISP connector to reset pin 9 of ATmega8515 microcontroller IC2 by shorting the jumper (JP).

Now start AVR Studio 4.0 without opening any project file. Optionally, you may proceed as follows: Main menu→Tools→Program AVR→Select AVR Programmer. Press 'Connect' after selecting STK500 or AVRISP in the platform window and COM port (say, COM1) in the port window.

After selection, 'AVR Studio STK500 Programming Menu' window appears. Select 'ATmega8515' as the device, 'ISP' as the programming mode and browse your project hex file (prog.hex) from Project→Debug→Exe folder.

You may verify fuse bits showing the default setting. Now press 'Program' button in 'Program' tab to burn the hex file into your microcontroller (IC1). Disconnect IC1 from the ISP cable. Now the programmer board is ready to burn the binary file into the target AVR microcontroller.

Software

The software (AVR910) for the microcontroller is written in 'C' using the IAR Embedded Workbench integrated development environment. IAR Embedded Workbench is being developed by IAR Systems and ATMEL developers in parallel and hence it generates the optimised code which uses full 'C' coding capabilities of AVR devices. AVR development tools for embedded systems can be